Chapter 3

Boolean Algebra & Logic gates

Introduction :-

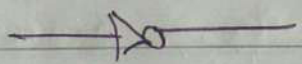
1850's - George Boole

Boolean algebra :-
It is used for design & analysis of digital system.

* Basic logical operation (Logic variables) :-
To simplify the logical expressions, used in digital ckt. we need to use logical operators.
three main logic operator.

1) NOT operator
2) AND operator
3) OR Operator.

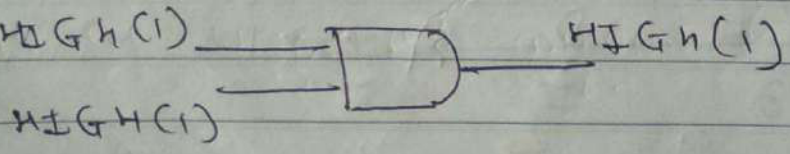
1) NOT operator



The NOT operator

$\overline{A}$ = NOT A

2) AND operator

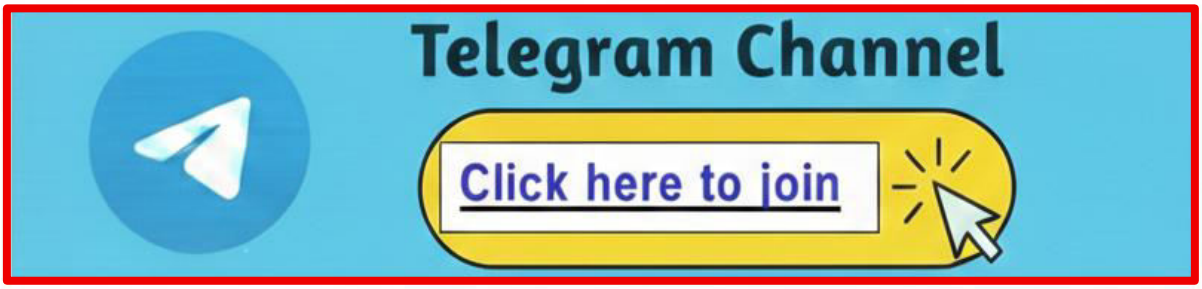


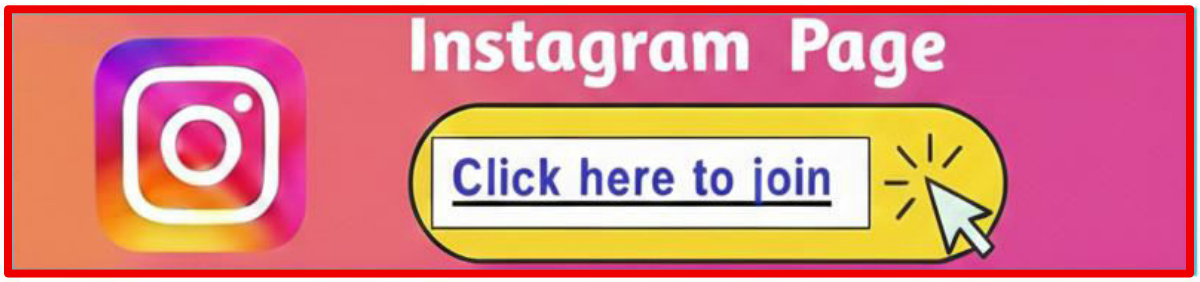
A.B - -- Logical multiplication

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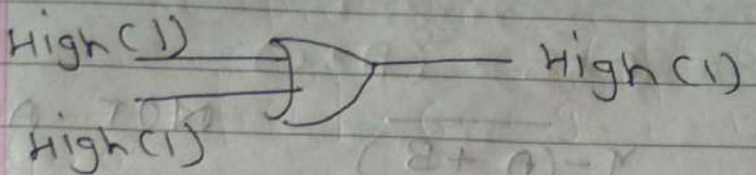




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OR operator :-



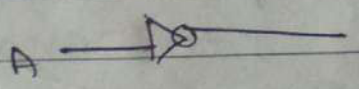
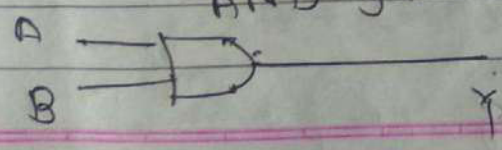
A+B — Logical Addition.

Logic gate :-
many i/p & one o/p.

Truth table :-
The operation of logic gate can be best understood with the help of table is called truth table.

Boolean algebra :-
The relation between the i/p & o/p of a gate can be expressed mathematically means of the boolean expression.

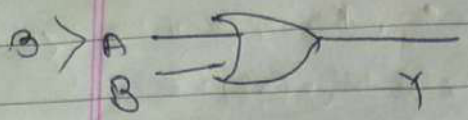
* Gate, symbol & Boolean expression :-
understand boolean algebra we need we of gate.

Sr. No.	Name of gate	Boolean expression	Logical operation.
1)		$Y = \overline{A}$	Inversion.
2)	AND gate 	$Y = A \cdot B$	Logical multiplication

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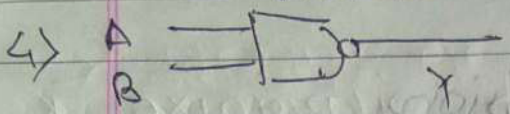
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3) OR gate



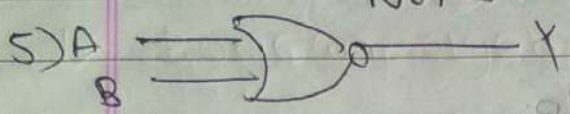
$Y = A + B$ Logical addition

4) NAND



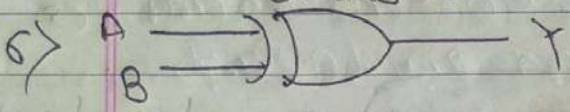
$Y = \overline{(A + B)}$ NOT AND

5) NOR



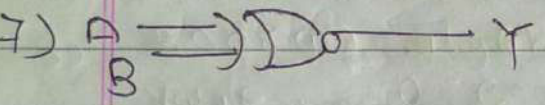
$Y = \overline{(A \cdot B)}$ NOT OR

6) Exclusive OR



$Y = A \oplus B$ Addition / substraction

7) Exclusive-NOR



$Y = \overline{A \oplus B}$ NOT-EXOR = EXNOR

A & B = i/p
Y = o/p

* Basic definitions :-

Boolean algebra can be defined with the help of following:

1) set of element
2) set of operator
3) No. of unproved axioms & postulates.

$A \in S$

$S = \{A, B, C, D\}$

$\cdot \rightarrow$ logical operator

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postulates :-

1) Deduce, rules, theorems & properties of system.
2) There are Imp postulate for the formulation of various algebraic structure
   1) closure
   2) Associative law
   3) commutative law
   4) Identity element
   5) Inverse
   6) Distributive law.

1) closure :-
   Set of element S is closed w.r.t to binary operator.

2) Associative Law :-

$$(A * B) * C = A * (B * C) \quad \text{for all } A, B, C \in S$$

3) commutative law

$$A * B = B * A \quad \text{for all } A, B \in S.$$

4) Identity element

$$x * A = A * x = A \quad \text{i.e. } x = 0, 1$$

$A \in S$.
$A \in I$.

$$A + 0 = 0 + A = A$$

5) Inverse :-

$$A * B = x$$

$$A + (-A) = 0$$

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6) Distributive law

*, · → Binary operator.

$A * (B \cdot C) = (A * B) \cdot (A * C)$

* Axiomatic definition of boolean algebra :

1) Boolean algebra → (Represent letter)
→ no numerical value

2) Arithmatic operation (addition, multiplication, substraction division) are not performed in boolean algebra. Also there is no fraction, -ve no., square, square root & logarithmic.

3) Boolean algebra allows only two possible values. ("0" & "1") (0 to 1) → for any variable

Rules for Boolean algebra

1) Binary 1 - high
Binary 0 - low

2) ANDing $A \cdot B \cdot C = A + B + C$

3) ORing $A \cdot B \cdot C = ABC$

4) $B = 0, \overline{B} = 1$; $B = 1, \overline{B} = 0$

Boolean postulate & law :

(+), (·) = Operator.

1) $A + 0 = 0 + A = A$

2) $A \cdot 1 = 1 \cdot A = A$

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3) $A + B = B + A$ } commutative law — w.r.t. 0 (+)

4) $A \cdot B = B \cdot A$ } commutative law — w.r.t. 0 (·)

5) Distributed over (+)

$$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$$

6) Distribu(·) over

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

7) $A + \overline{A} = 1$ & $A \cdot \overline{A} = 0$

* Two valued boolean algebra :-

I/P A	I/P B	O/P A·B	I/P A	I/P B	O/P A+B
0	0	0	0	0	0
0	1	0	0	1	1
1	0	0	1	0	1
1	1	1	1	1	1

(a) AND operator. (b) OR operator

AND, OR, NOT

I/P A	O/P $\overline{A}$
0	1
1	0

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1) Closure :-

AND, OR, NOT $\longrightarrow$ 0 or 1

$0, 1 \in S$.

2) Identity element

$0 + 0 = 0$

$0 + 1 = 1 + 0 = 1$

$1 \cdot 1 = 1$

$1 \cdot 0 = 0 \cdot 1 = 0$.

3) Commutative law

$A \cdot B = B \cdot A$

& $A + B = B + A$

I/P

A	B	$A \cdot B$	$B \cdot A$	$A + B$	$B + A$
0	0	0	0	0	0
0	1	0	0	1	1
1	0	0	0	1	1
1	1	1	1	1	1

$A \cdot B = B \cdot A$ $\quad$ $A + B = B + A$

4) Distributive law.

$A \cdot (B + C) = AB + AC$

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Boolean law for simplification of ex,

$= A + AB$

5) Inverse :-

$A + \overline{A} = 1$ $\quad A \cdot \overline{A} = 0$

i.e. $0 + 1 = 1$ $\quad 0 \cdot 1 = 0$

$1 + 0 = 1$ $\quad 1 \cdot 0 = 0$

6) Associative law :-

$(A \cdot B) \cdot C = A \cdot (B \cdot C)$

$(A + B) + C = A + (B + C)$

* Basic theorem & properties of Boolean algebra :-

1) Duality theorem :-

Duality theorem following conversion are possible.

1) change each ~~AND~~ operation to an OR operation

2) change each OR operation to an AND operation.

3) complement any 1 or 0 appearing in the expression.

It is useful in creating new expression from given expression

① $A + 1 = 1$

$\Rightarrow$

$A \cdot 0 = 0$

② $A(B + C) = AB + BC$

$A + (BC) = (A + B) \cdot (A + C)$

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Basic theorems :-

1) AND law

A = 1st i/p

i) $A \cdot 0 = 0$ 0 = 2nd i/p

ii) $A \cdot 1 = A$

iii) $A \cdot A = A$

iv) $A \cdot \overline{A} = 0$

2) OR Laws.

i) $A + 0 = A$

ii) $A + 1 = 1$

iii) $A + A = A$

iv) $A + \overline{A} = 1$

3) Inversion law

$A = \overline{\overline{A}}$

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Inversion law :-

Boolean laws.

Sr. NO.	Name	Statement of Law
1)	Commutative Law	$A \cdot B = B \cdot A$ $A + B = B + A$
2)	Associative law	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$ $(A + B) + C = A + (B + C)$
3)	Distributive law	$A \cdot (B + C) = AB + AC$
4)	AND Law	$A \cdot 0 = 0$ $A \cdot 1 = A$ $A \cdot A = A$ $A \cdot \overline{A} = 0$
5)	OR Law	$A + 0 = A$ $A + 1 = 1$ $A + A = A$ $A + \overline{A} = 1$
6)	Inversion law	$\overline{\overline{A}} = A$

7) Other Imp law

1) $A + BC = (A + B)(A + C)$
2) $\overline{A} + AB = \overline{A} + B$
3) $\overline{A} + A\overline{B} = \overline{A} + \overline{B}$
4) $A + AB = A$
5) $A + \overline{A}B = A + B$

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* De-Morgan's Theorems :-
Two theorem

Theorem (I) :-

1) $\overline{AB} = \overline{A} + \overline{B}$

NAND = Bubbled OR

⇒ This suggests that the complement of product is equal to addition of the complements.

NAND = Bubbled OR

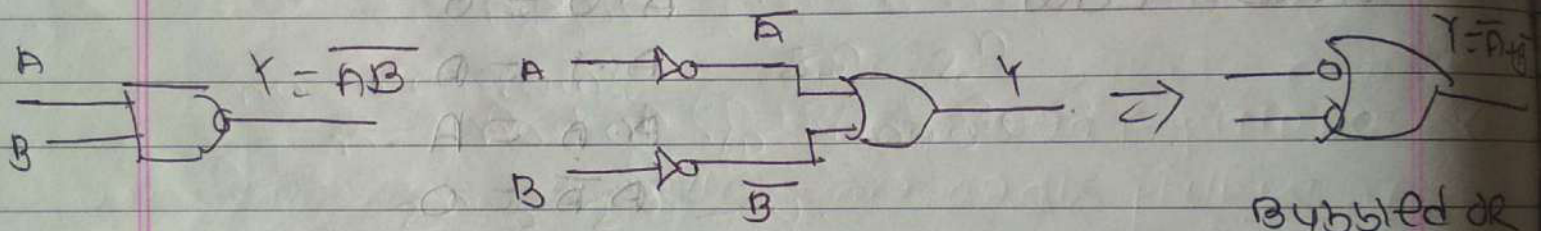


Illustration of demorgan's first theorem

A	B	$\overline{AB}$	$\overline{A}$	$\overline{B}$	$\overline{A}+\overline{B}$
0	0	1	1	1	1
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	0	0	0

$AB = \overline{A}+\overline{B}$

Theorem (II) $\overline{A+B} = \overline{A} \cdot \overline{B}$:- NOR = Bubbled AND

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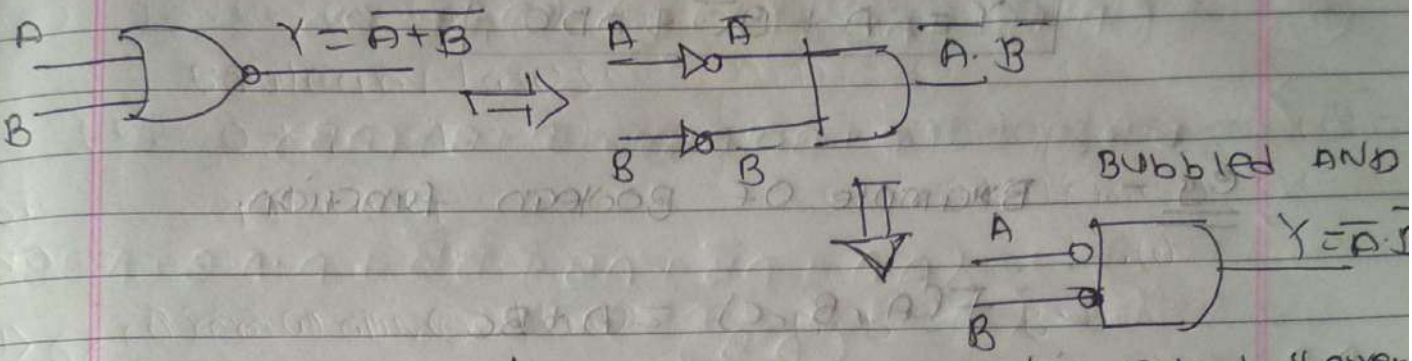


Illustration of De-morgan's second theorem

A	B	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cdot \overline{B}$
0	0	1	1	1	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	0

$\overline{A+B} = \overline{A} \cdot \overline{B}$

fig :- Truth table to verify De-Morgan's theorem.

* **Operator precedance :-**

operator precedence for solving expression

1) parenthesis
2) NOT
3) AND
4) OR

* **Boolean Expression & Boolean function :-**

$F(A, B, C, D) = A + B\overline{C} + AD\overline{C}$

Boolean function — Boolean expression

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$$Y = A + B\bar{C} + ADC$$

eg :- Example of Boolean function.

$$f(A, B, C) = A + BC$$

or simply $Y = A + BC$

Truth table formation :-

It is possible to convert switching eqn in truth table

$$f(A, B, C) = A + BC$$

no. of rows = 2^n

n = no. of i/p variable

$2^n = 2^3 = 8$.

I/p			O/p
A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

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Example on reducing Boolean expression:-

1) Prove the following boolean expression:

i) $A + AB = A$

ii) $A + \bar{A}B = AB + B$

iii) $(A+B)(A+C) = A + BC$

⇒ i) $A + AB = A$

L.H.S $= A + AB = A(1+B)$ $\because 1+B = 1$

$= A$

$\boxed{A + AB = A}$

ii) $= A + \bar{A}B$ since $A = A + AB$

$= A + AB + \bar{A}B$

$= A + B(1)$

$= A + B$

$\boxed{A + \bar{A}B = A + B}$

iii) $(A+B)(A+C) = A + BC$

L.H.S $= AA + AC + BA + BC$ $A \cdot A = A$

$= A(1 + C) + BA + BC$

$= A(1+B) + BC$

$\boxed{(A+B)(A+C) = A + BC}$

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(2) prove that

$(A+\bar{B}+AB)(A+\bar{B})(\bar{A}B) = 0$

$\Rightarrow$ L.H.S. $= (A+\bar{B}+AB)(A+\bar{B})(\bar{A}B)$

$= (AA + A\bar{B} + A\bar{B} + \bar{B}\bar{B} + A\cdot A\cdot B + A\cdot B\bar{B})(\bar{A}B)$

$A\cdot A$

$= (A + \bar{B}(A+A) + \bar{B} + AB + A\cdot 0)(\bar{A}B)$

$= (A + \bar{B}\cdot A + \bar{B} + AB + 0)(\bar{A}B)$

$= A\bar{A}B + \bar{A}AB + \bar{B}B\bar{B} + A\bar{A}BB$

$= 0 + \bar{A}(A+B) + 0 \quad 0+0+0+0$

$= 0$

$$\boxed{(A+\bar{B}+AB)(A+\bar{B})(\bar{A}B) = 0}$$

(3) Show that

$A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C = B + C$

$\Rightarrow$ L.H.S $= A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C$

$= C(A\bar{B} + \bar{A}) + B(1 + \bar{D} + D\bar{D} + AB)$

$\bar{A}\cdot(1+\bar{B})$ $= C\cdot(A\bar{B} + \bar{A}) + B \times 1$

$= ((\bar{A}(1+\bar{B}) + A\bar{B}) + B$

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$B + \bar{B}C = B + C$

$= C \cdot [\bar{A} + \bar{A}B + \bar{A}\bar{B}] + B$

$= C[\bar{A}(1 + B) + \bar{A}\bar{B}] + B$

$= C[\bar{A} + \bar{B}] + B$

$= B + C$

$= \bar{A}C + \bar{B}C + B$

$= \bar{A}C + B$

$= \bar{A}C + B + C = C + B$

$\boxed{A\bar{B}C + B + B\bar{D} + AB\bar{D} + \bar{A}C = B + C}$

(4) Simplify following expression with boolean laws.

→

i) $F = XY + XYZ + XY\bar{Z} + X\bar{Z}Y$

⇒

$= XY(1 + Z) + XY(Z + \bar{Z})$

$= XY + XY(1)$

$\boxed{F = XY}$

ii) $F = \bar{A}B(\bar{D} + D\bar{C}) + (A + DAC)\bar{B}$

$\bar{D} + D\bar{C} = \bar{D} + \bar{C}$

$= \bar{A}B(\bar{D} + \bar{C}) + A\bar{B}(1 + CD)$

$\boxed{F = \bar{A}B\bar{D} + \bar{A}B\bar{C} + A\bar{B}}$

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⑤ simplify following expression

$$Y = \overline{(\overline{AB} + \overline{A} + AB)}$$

$\Rightarrow$

$$Y = \overline{(\overline{AB} + \overline{A} + AB)}$$

$$\overline{AB} = \overline{A} + \overline{B}$$

$$= \overline{(\overline{A} + \overline{B} + \overline{A} + AB)}$$

$$\overline{A} + \overline{A} = \overline{A}$$

$$= \overline{(\overline{A} + \overline{B} + AB)}$$

$$\overline{A + B + C} = \overline{A} \cdot \overline{B} \cdot \overline{C}$$

$$= \overline{\overline{A}} \cdot \overline{\overline{B}} \cdot \overline{AB}$$

$$= A \cdot B \cdot \overline{AB}$$

$$= A \cdot B \, (\overline{A} + \overline{B})$$

$$= A \cdot \overline{A} B + B \cdot \overline{B} A$$

$$\boxed{Y = 0}$$

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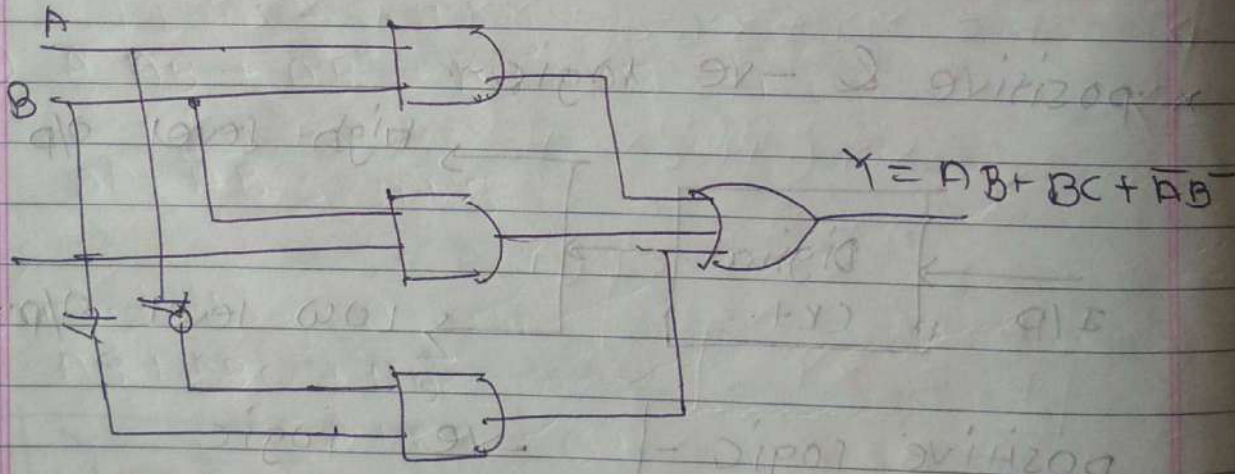
1) ~~NOT gate~~ ~~IC NO~~ :

Gate	IC NO
1) NOT	7404
2) AND	7408
3) OR	7432
4) NAND	7400
5) NOR	7402
6) EX-OR	7486
7) EX-NOR	74266

* Draw logic ckt from Boolean Equation:

1) $Y = AB + BC + \overline{A}\overline{B}$

⇒



2) Sketch the given boolean expression using one AND gate one OR gate only.

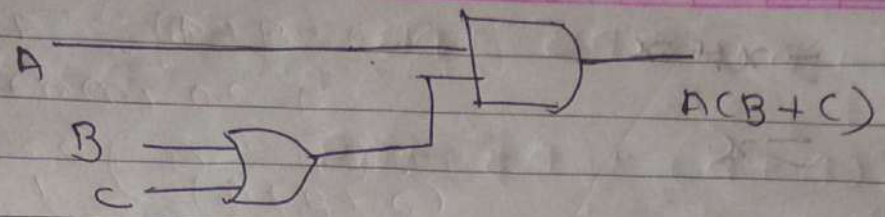
$Y = AB + BC$

⇒

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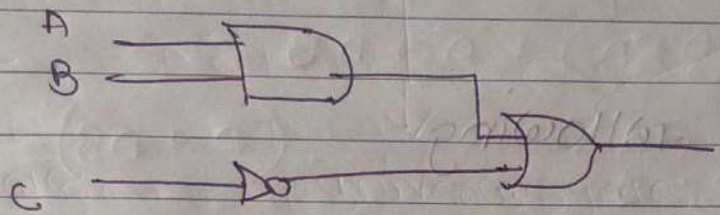
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i) $Y = A(B+C)$



* write boolean expression from logic ckt.

1) obtain boolean eqn for logic ckt.



$\Rightarrow$ $\boxed{Y = (AB) + \overline{C}}$

* Examples –

1) Apply Boolean algebra and minimize the following functions.

i) $xy + \overline{x}z + yz$

ii) $(x+y)(x+\overline{y})$

iii) $x(\overline{x}+y)$

$\Rightarrow$

i) $xy + \overline{x}z + yz$

$\rightarrow$ $y(x+z) + \overline{x}z$

ii) $x \cdot x + x \cdot \overline{y} + xy + y \cdot \overline{y}$

$= x + xy + 0$

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$= x + x$

$= x$

iii) $x(\bar{x}+y) = x \cdot \bar{x} + xy$

$= 0 + xy$

$= xy$

(2) prove the following.

i) $AB + ABC + A\bar{B} = A$

ii) $(B+A)(B+D)(A+C)(C+D) = BC + AD$

iii) $(\bar{A}+B)\bar{A}\bar{B}\bar{C} = \overline{A+B+C}$

iv) $\bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB = 1$

v) $A + \bar{A}B = A + B$

$\Rightarrow$ L.H.S. =

i) $AB + ABC + A\bar{B}$

$= AB(1+C) + A\bar{B}$

$= AB + A\bar{B}$

$= A(B + \bar{B})$ $\quad (B + \bar{B}) = 1$

$= A$

L.H.S. = R.H.S.

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L·H·S·

ii) $= (B+A)(B+D)(A+C)(C+D)$

$= (B \cdot B + B \cdot D + AB + AD)(AC + AD + CC + CD)$

$= (B + BD + AB + AD)(AC + AD + C + CD)$

$= B(1 + D + A) + AD)(A(C + D) + C(1 + D))$

$= (B + AD)(A(C + D) + C)$

$= (B + AD)(AC + AD + C)$

$= (B + AD)(C + AD)$

$= BC + ABD + ADC + AD \cdot AD$

$= BC + AD(B + C) + AD$

$= BC + AD(B + C + 1)$

$= BC + AD$

L·H·S = R·H·S

iii) $L \cdot H \cdot S = (\overline{A}+B)\overline{D}\,\overline{B}\,\overline{C}$

$= \overline{A} \cdot \overline{A}\,\overline{B} \cdot \overline{C} + B \cdot \overline{B}\,\overline{A}\,\overline{C}$

$= \overline{A}\,\overline{B}\,\overline{C} + 0$

$= \overline{A + B + C}$

L·H·S = R·H·S

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iv) $= A\bar{B} + \bar{A}\bar{B} + \overline{AB} + AB$

$= \bar{B}(A + \bar{A}) + \overline{AB} + AB$

$= (\bar{B} + \bar{A} + B) + AB$

$= 1 \quad \overline{A \cdot B \cdot C} + AB$

$= A\bar{B} + \bar{A}\bar{B} + \bar{A} + \bar{B} + AB$

$= \bar{B}(1 + A + \bar{A}) + \bar{A} + AB$

$= \bar{B} + \bar{A} + AB \qquad \therefore (\bar{A} + AB) = (\bar{A} + B)$

$= \bar{B} + \bar{A} + B \qquad (\because B + \bar{B} = 1)$

$= 1 + \bar{A} \qquad \therefore (1 + \bar{A} = 1)$

L.H.S = 1

L.H.S = R.H.S.

v) $A + \bar{A}B = A + B$

$\Rightarrow$ L.H.S = $A + \bar{A}B \qquad (\because A = A + AB)$

$= A + AB + \bar{A}B$

$= A + B(A + \bar{A})$

$= A + B$

L.H.S = R.H.S.

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Universality of NAND gate:-

NAND gate expressed mathematically:-
6 convert each logic function into an expression using NAND gate only.

$$Y = \overline{A \cdot B}$$

1) NOT gate (Inverter) using NAND gate:-

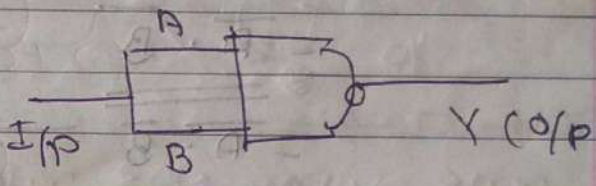
I/P $= A = B = A$

$$Y = \overline{A \cdot B}$$

$$= \overline{A \cdot A}$$

$$Y = \overline{A}$$

$A \cdot A = A$

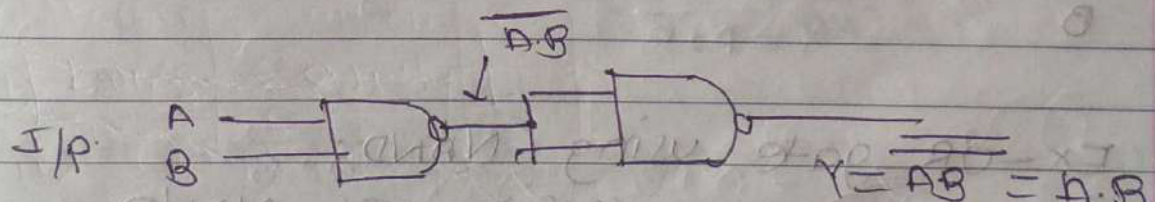


NOT using NAND

2) AND gate using NAND:-
Boolean expression for AND.

$$Y = \overline{\overline{A \cdot B}}$$

$$Y = \overline{\overline{AB}} = A \cdot B$$



3) OR gate using NAND gate

$$Y = A + B$$

$$Y = \overline{\overline{A + B}}$$

$$= \overline{\overline{A} \cdot \overline{B}}$$

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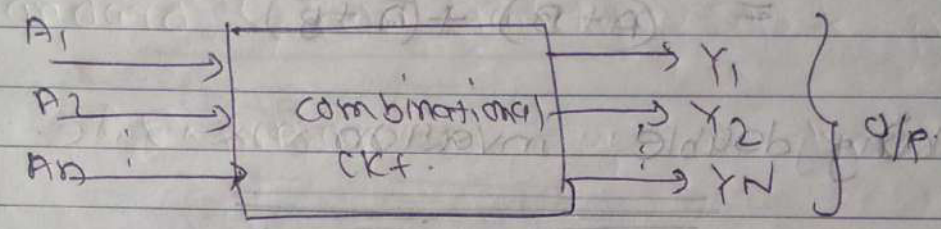
Logic Minimization & Reduction Techniques-

PAGE No.
DATE

Intd to combinational ckt.

Types of digital system
1) ~~Analog~~ Combinational logic ckt.
2) sequential logic ckt.

combinational ckt :-



n = i/p
m = o/p

e.g of combinational ckt :-
1) Adder, substractor
2) code converter
3) Encoder, decoder.
4) comparator
5) Multiplexer, Demultiplexer.

Approaches of combinational ckt. design :-
1) Traditional method
2) use of MSI (medium scale integration)

Method to simplify boolean function
1) Algebric method
2) Karnaugh-map Simplification
3) Quince-Mc Cluskey method
4) VEM (variable entered mapping) technique

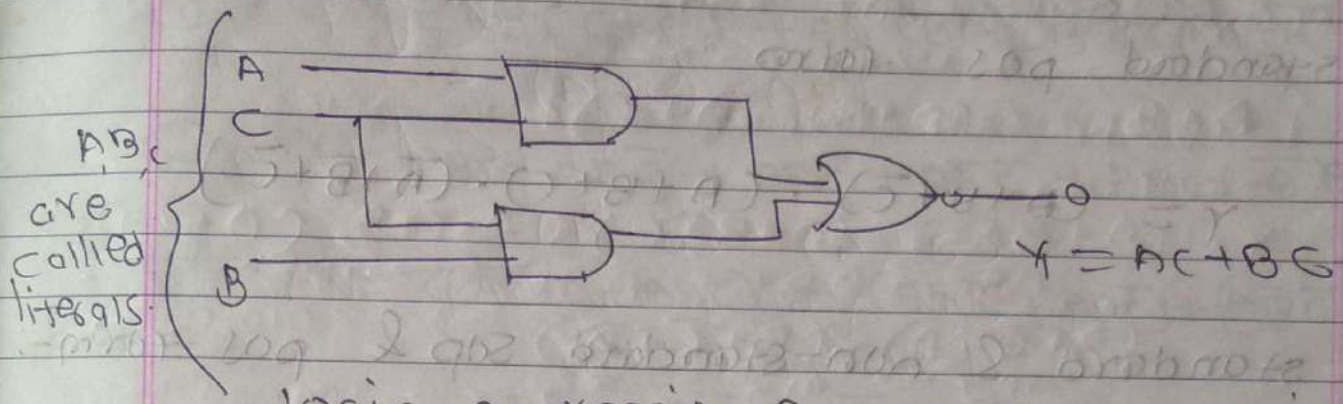
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* SOP & POS Representation for logic expressions :-

Logic expression

$Y = AC + BC$



logic expression can be expressed into following two standard format.

1) Sum of product (SOP) &
2) product of sum (POS) form.

* Sum of product (SOP) forms :-

$Y = AB + AC + BC$

sum — literals are ANDed together.

product terms.

* product of sum (POS) form

$Y = (A+B) \cdot (B+C) \cdot (A+C)$

literals are ORed together.

* Standard or canonical SOP & POS form - 8

canonical - confirming general rule

two canonical form

1) SOP format
2) POS format

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each sum

standard SOP form:-

$$Y = ABC + A\bar{B}\bar{C} + \bar{A}BC$$

standard POS form

$$Y = (A+B+\bar{C}) \cdot (A+B+C) \cdot (\bar{A}+\bar{B}+\bar{C})$$

standard & non-standard SOP & POS form:-

Sr. No.	Expression	type
1)	$Y = AB + A\bar{B}\bar{C} + ABC$	Non-standard SOP
2)	$Y = AB + A\bar{B}C + \bar{A}\bar{B}$	standard SOP
3)	$Y = (\bar{A}+B) \cdot (A+B) \cdot (A+\bar{B})$	standard POS
4)	$Y = (\bar{A}+B) \cdot (A+B+C)$	Non-standard POS

* conversion of a logic expression to standard SOP or POS form:-

convert from SOP to standard SOP form

Step ① - For each term find the missing literal

② The AND term with the term formed by ORing the missing literal & it's complement

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e.g convert the exp into standard SOP form.

$Y = AB + A\bar{C} + BC$

$\Rightarrow$

① $Y = AB\cdot(C+\bar{C}) + A\bar{C}(B+\bar{B}) + BC(A+\bar{A})$

② $Y = ABC + AB\bar{C} + AB\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + \bar{A}BC$

$= (ABC + ABC) + (AB\bar{C} + AB\bar{C}) + (A\bar{B}\bar{C} + \bar{A}BC)$

$A + A = A$

$= ABC + AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC$

↑ standard SOP form

conversion from standard POS to standard POS form.

ANDing the missing literal & complement.

e.g convert the expression $Y = (A+B)(A+C)(B+\bar{C})$ into standard POS form :-

$\Rightarrow$

$Y = (A+B+C\bar{C})(A+C)(B\cdot\bar{B})(B+\bar{C})(A\cdot\bar{A})$

$= (AC\bar{C} + BC\bar{C})(AB\bar{B} + BC\bar{B})(B\bar{A}\bar{A} + \bar{A}AC)$

$= (A\cdot A\cdot B\cdot\bar{B}\cdot C\bar{C} + A\cdot C\cdot\bar{C}B\cdot C\cdot\bar{B} + B\cdot C\cdot\bar{C}\cdot A\cdot B\bar{B} + B\cdot C\cdot\bar{B}\cdot B C\bar{C})\cdot(B\bar{A}\bar{A} + \bar{A}AC)$

$= (A + B + C\bar{C})(A + B\bar{B} + C)(A\bar{A} + B + \bar{C})$

$= (A\cdot A\cdot B\cdot\bar{B} + A\cdot C + A\cdot B + B\cdot B\cdot\bar{B} + BC)(A\bar{A} + B + \bar{C})$

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$= (A + 0 + AC + AB + BC)(A\bar{A} + B + \bar{C})$

$= A(1 + C + B)(A\bar{A} + B + \bar{C})$

$\boxed{A + BC = (A + B)(A + C)}$

$\therefore Y = (A + B + C)(A + B + \bar{C})(A + C + B)(A + C + \bar{B})$
$(B + \bar{C} + A)(B + \bar{C} + \bar{A})$

$A \cdot A = 1$

$= (A + B + C)(B + \bar{C} + A)(A + C + B)(B + \bar{C} + \bar{A})$
$(A + C + \bar{B})$

Standard POS form.

* Concept of minterm & Maxterm :-

Minterm :-
Each individual term in standard SOP form is called as minterm.

Maxterm :-
Each individual term in the standard POS form is called a maxterm.

Standard SOP

$Y = ABC + A\bar{B}\bar{C} + \bar{A}B\bar{C}$

each individual term is called minterm.

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Standard POS :-

$X = (A+B) \cdot (A+\bar{B})$

each individual term called maxterm.

$2^3 = 8$ 3 = variable

fig :- Minterm & Maxterm for 3 variable

Variable			Minterms	Maxterms
A	B	C	m_i	M_i
0	0	0	$m_0 = \bar{A}\bar{B}\bar{C}$	$M_0 = A+B+C$
0	0	1	$m_1 = \bar{A}\bar{B}C$	$M_1 = A+B+\bar{C}$
0	1	0	$m_2 = \bar{A}B\bar{C}$	$M_2 = \bar{A}+\bar{B}+C$
0	1	1	$m_3 = \bar{A}BC$	$M_3 = A+\bar{B}+\bar{C}$
1	0	0	$m_4 = A\bar{B}\bar{C}$	$M_4 = \bar{A}+B+C$
1	0	1	$m_5 = A\bar{B}C$	$M_5 = \bar{A}+B+\bar{C}$
1	1	0	$m_6 = AB\bar{C}$	$M_6 = \bar{A}+\bar{B}+C$
1	1	1	$m_7 = ABC$	$M_7 = \bar{A}+\bar{B}+\bar{C}$

$m_i = i = 0, 1, 2 \dots 2^n - 1$

how minterm & maxterm obtained.

minterm Let $ABC = 011$

A, B, C i/p to and gate we want o/p of and 1

invert 0 minterm = $\bar{A}BC$ o/p = 1

Maxterm $ABC = 011$

A, B, C i/p to or gate o/p = 0

invert 1

maxterm = $(A+\bar{B}+\bar{C})$

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Σm → minterm $N \leq (2^n) - 1$

πM → Maxterm

Representation of logical expression using minterm & Maxterm :-

Represent logical expression using minterm & maxterm.

① $Y = \underbrace{ABC}_{m_7} + \underbrace{\bar{A}BC}_{m_3} + \underbrace{A\bar{B}\bar{C}}_{m_4}$ ← Given logical expression

← corresponding minterm

$\therefore Y = m_7 + m_3 + m_4$

$Y = \Sigma m\,(3, 4, 7)$ → other wave of representation

Σ denotes sum of products

② $Y = \underbrace{(A + \bar{B} + C)}_{M_2} \cdot \underbrace{(A + B + C)}_{M_0} \cdot \underbrace{(\bar{A} + \bar{B} + C)}_{M_6}$ ← G.E

$Y = M_0 M_2 M_6$

$Y = \pi M\,(0, 2, 6)$ ← other way of Repre

π denotes product of sum.

Simplification technique for Boolean expres

1) Algebric simplification
2) simplification using karnaugh maps.
3) Quine MC-cluskey method
4) variable Entered Mapping (VEM) method.

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Algebric simplification:-

1) Bring the given expression in SOP form by using boolean law & De-Morgan's theorem.
2) simplify the SOP expression by checking product terms for common factor.

e.g ① Simplify the expression

$$Y = AB + (A+B)\cdot(\overline{A}+B)$$

$\Rightarrow$

$$= AB + (A\overline{A} + AB + \overline{A}B + B\cdot B)$$

$$= AB + (0 + 1 + AB + \overline{A}B)$$

$$= AB + B(1 + B(A+\overline{A}))$$

$$= AB$$

$$= AB + 1 + AB + \overline{A}B$$

$$= AB + 1 + \overline{A}B$$

$$= B(A+\overline{A}) + 1$$

$$= B + 1$$

$$= AB + A\cdot\overline{A} + AB + \overline{A}B + B\cdot B$$

$$= AB + 0 + \overline{A}B + B$$

$$= AB + B + \overline{A}B$$

$$= B(1+A) + \overline{A}B \qquad A+1 = 1$$

$$= B + \overline{A}B$$

$$= B(1+\overline{A}) \qquad 1+\overline{A} = 1$$

$$\boxed{Y = B}$$

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② simplify three variable boolean expression

$Y = \Sigma m(2,4,6)$

$\Rightarrow$ $Y = m_2 + m_4 + m_6$

$= \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$

$= \bar{A}B\bar{C} + A\bar{C}(\bar{B}+B)$

$= \bar{A}B\bar{C} + A\bar{C}$

$N \leq 2^n - 1$

$N \leq (2^3) - 1$

$= \bar{C}(\bar{A}B + A)$

$= \bar{C}[(\bar{A}+A)(A+B)]$

$= \bar{C}\cdot(A+\bar{A})\cdot(A+B)$ $\quad A+\bar{A} = 1$

$\boxed{Y = (A+B)\cdot\bar{C}}$

③ simplify the following three variable logic expression

$Y = \pi m(1,3,5)$

$\Rightarrow$ $Y = M_1 \cdot M_3 \cdot M_5$

$Y = (A+B+\bar{C})(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})$

$= (A\cdot A + A\bar{B} + A\bar{C} + BA + B\bar{B} + B\bar{C} + A\bar{C} + \bar{B}\bar{C} + \bar{C}\cdot\bar{C})$

$= (A + A\bar{B} + A\bar{C} + AB + B\bar{C} + A\bar{C} + \bar{B}\bar{C} + \bar{C}$

$(\bar{A}+B+\bar{C})$

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$= (A + A\bar{B} + A\bar{C} + AB + B\bar{C} + \bar{B}\cdot\bar{C} + \bar{C})(\overline{A+B+C})$

$= (A + A\bar{C} + AB + B\bar{C} + \bar{B}\bar{C} + \bar{C})(\bar{A} + B + C)$

$= (A\cdot(1 + \bar{C} + B) + \bar{C}(1 + B + \bar{B}))(\bar{A} + B + C)$

$= (B + \bar{C})(\bar{A} + B + \bar{C})$

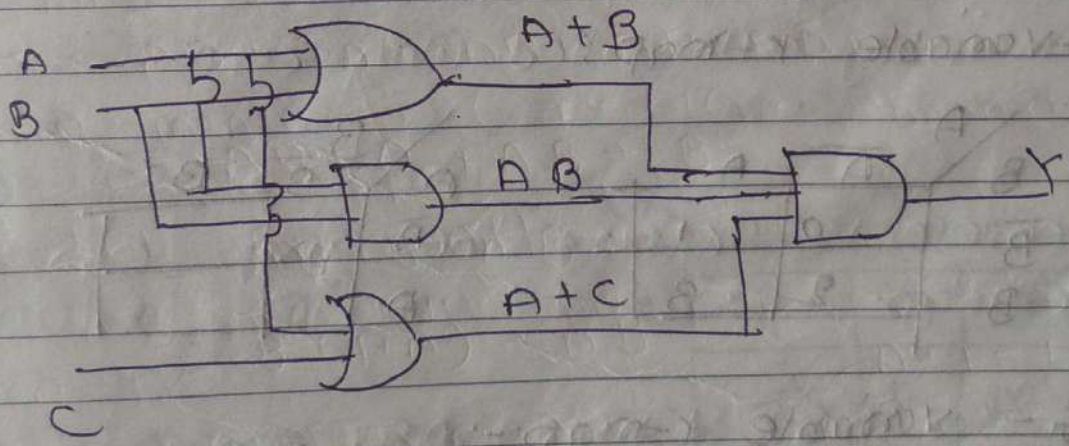
$= A\bar{A} + A\cdot B + A\bar{C} + \bar{A}\bar{C} + B\bar{C} + \bar{C}\cdot\bar{C}$

$= 0 + AB + A\bar{C} + \bar{A}\bar{C} + B\bar{C} + \bar{C}$

$= AB + \bar{C}(A + \bar{A}) + B + 1)$

$$\boxed{Y = AB + \bar{C}}$$

(3) For logic ckt write boolean exp. & simplify it



$Y = (A+B)\cdot AB\cdot(A+C)$

$= (A\cdot B\cdot A + AB\cdot C)(A+B)$

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as per gray or Karnaugh — Graphical Meth

k-map - Ben

$= (AAB + ABB + AABC + ABBC)$

$= \underline{AB + AB} + \underline{ABC + AB\bar{C}}$

$= AB + ABC$

$= AB(1 + C)$

$\boxed{Y = AB}$

K-map simplification :-

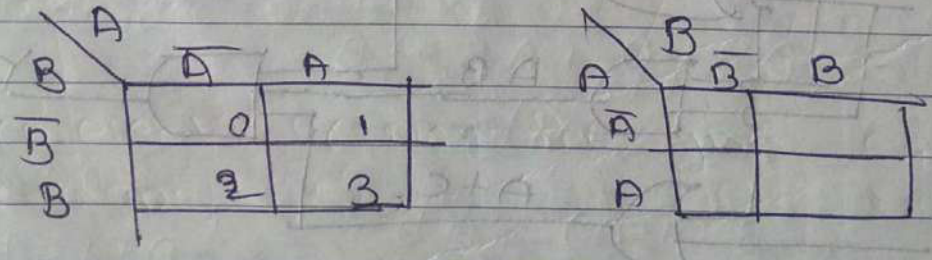
$\Rightarrow 2^n = 2^2 = 4$ boxes n = no. of variable

$2^3 = 3$ variable $= 8$ boxes.

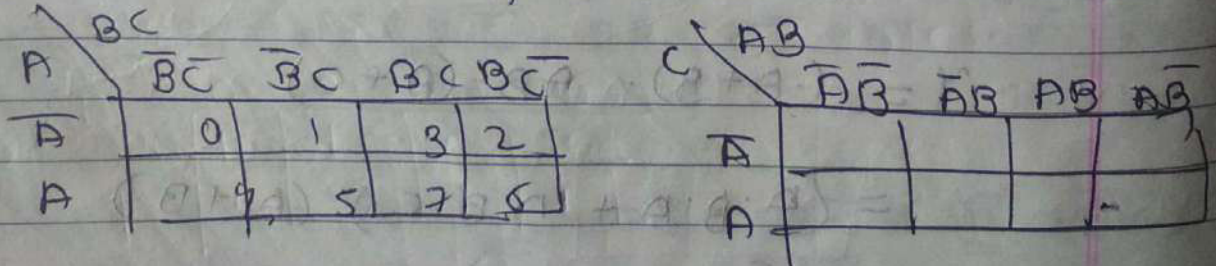
$2^4 = 4$ variable $= 16$ boxes.

0's & 1's represent the value of corresponding var...

2-variable k-map :-



3- variable k-map :-



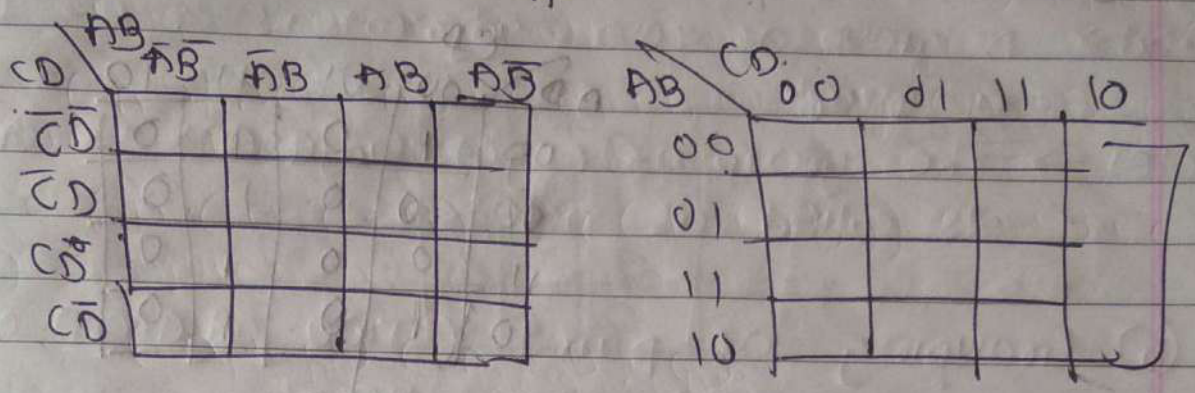
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pair = adjacent 2
quad = " 4
octet = " 8

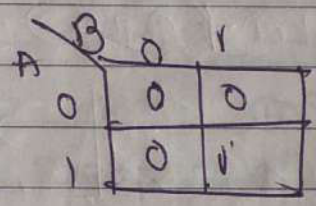
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4-variable k-map:



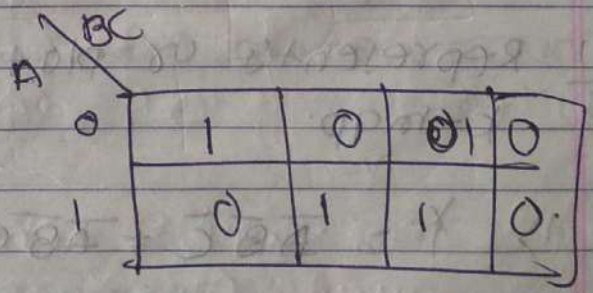
* k-map associated with product term:

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1



2-variable

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



3-variable.

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A B C D Y

AB \ CD	00	01	11	10
00	1	0	1	0
01	0	0	1	0
11	0	0	1	0
10	1	0	1	0

OR

CD \ AB				
	0	4	12	8
	1	5	13	9
	3	7	15	11
	2	6	14	10

* Represent standard SOP form on K-map :-

1) Simply entering 1's in a cell of the K-map corresponding to each minterm present in the equation.
2) Remaining cell filled with zero.

Q.6.) Represent's or plot the given eqn on K-map.

1) $Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C} + ABC$

⇒

A \ BC	00	01	11	10
0	1	1	0	0
1	1	0	1	1

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$Y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}C\bar{D} + AB\bar{C}\bar{D}$

AB \ CD	00	01	11	10
00	1	0	1	0
01	0	0	0	1
11	1	0	0	0
10	0	0	0	0

CD \ AB

Representation of standard SOP & K-map.

* simplification of Boolean expression using K-map -

Adjesont cell :- They differ by only one variable

left cell & Right cell

Top cell & Bottom cell

Simplification :-

Grouping means combining the term in adjecent cell.

Grouping of adjecent '0' — SOP form

adjecent '1' — POS form

only SOP. So we will consider group '1'. not '0'.

way of grouping :-

1) group most no. of 1's (or 0's)
2) Grouping follows Binary rules i.e. we can group 1, 2, 4, 8, 16, 32 -- number of 1's & 0's.
3) we can't group [illegible] - number of

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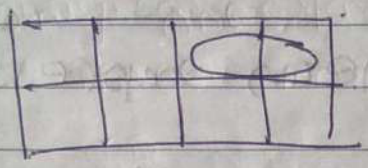
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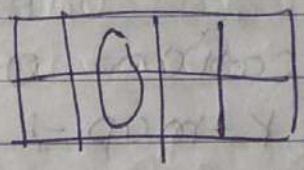
pair → A group of two adjacent 1's or 0's is called as a pair.

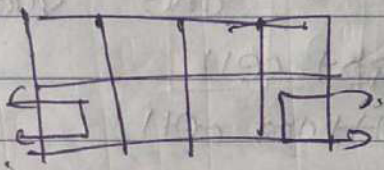
quad → A group of four adjacent 1's or 0's is called a quad.

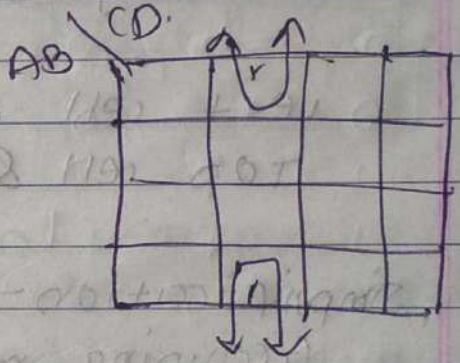
octet → The group of eight adjacent 1's or 0's is called as octet.

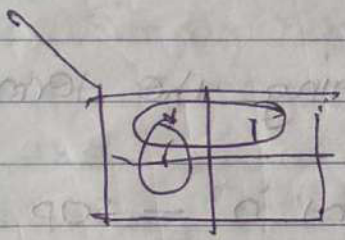
(a) Grouping of pairs (Two adjacent one's) pair - missing one literal.

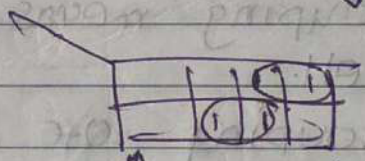












(ii) overlapping eliminate 2 variables.
2 pairs are sufficient to include all 1's.

* Grouping of four adjacent ones (quad

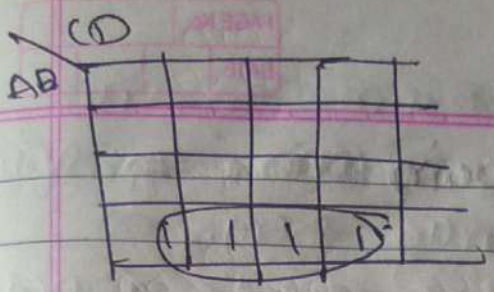
(iii) Two variables which are not same elimin
quad eliminates 2 variables.

1) Four horizontal adjacent ones.

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four vertical adjacent one's

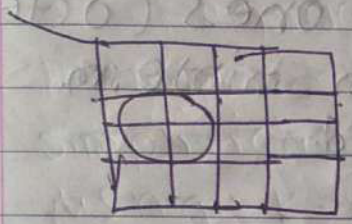




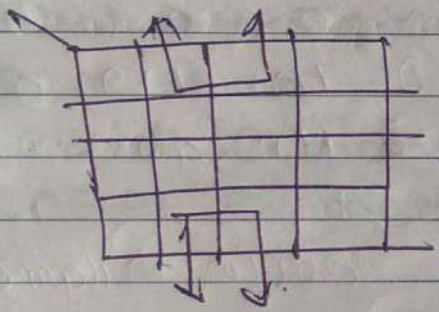
Two variables are same in all minterms.

The variable will not same eliminate.

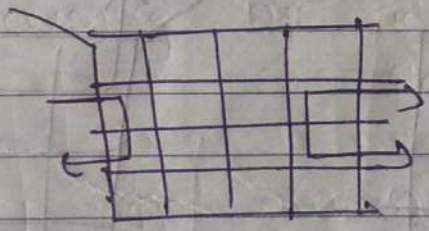
four adjacent 1's forming a square.



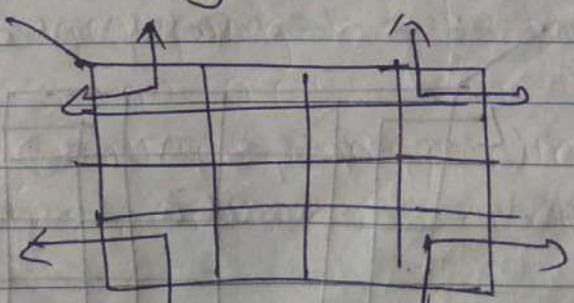
Top & bottom forming quad.



Leftmost & rightmost forming a quad



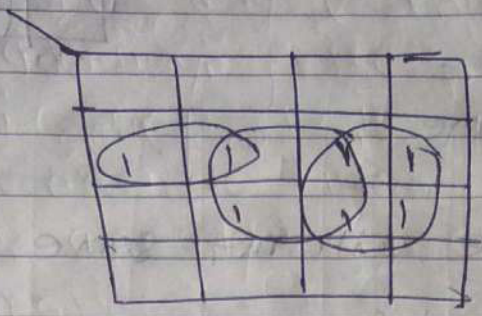
1's corresponding to corner forming a quad



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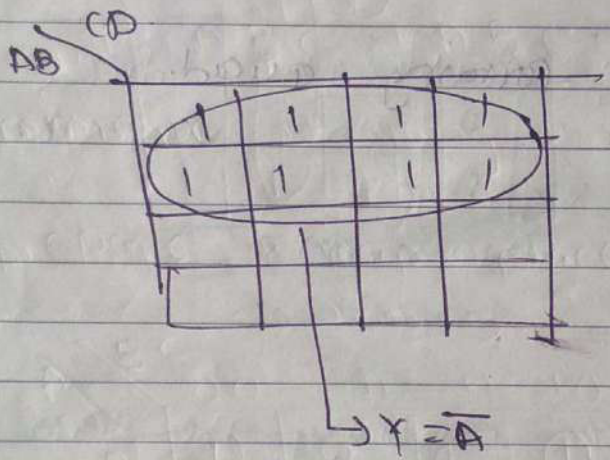
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overlapping quad & pair

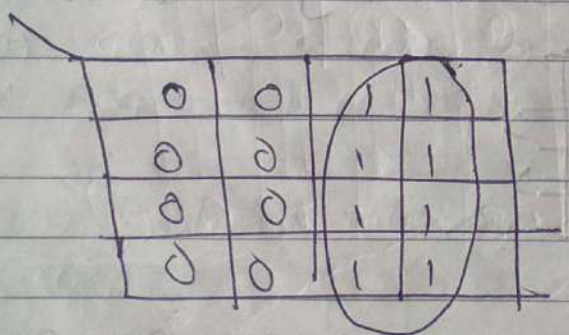


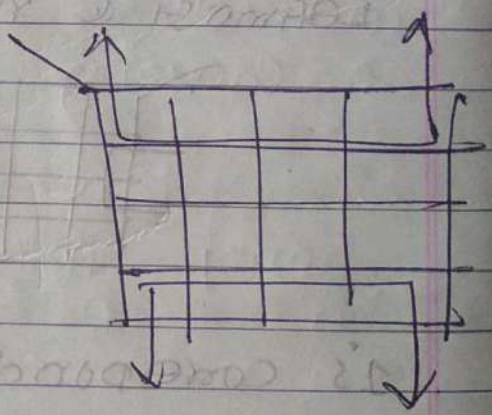
* Grouping Eight adjacent ones (octet):-
only one variable same in minterm.
Three variables eliminated.

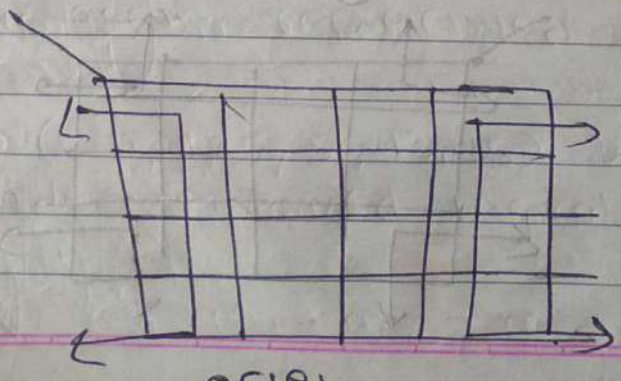
①



②







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Rules followed for k-map simplification (SOP)

1) Group may not contain include any cell containing a zero.
2) Group may be horizontal, vertical but not diagonal.
3) Group must contain 1, 2, 4, 8 or in general 2^n cells. Identify largest possible group first. $n=2 \Rightarrow n^2=4$
4) Each group should be as large as possible
5) Each cell containing a one must be at least in one group.
6) Group may overlap.
7) wrap around allowed.
8) Few groups are possible as long as this does not contradict any of the previous rule.
9) pair eliminates ~~three~~ variable.
10) quad eliminates two variables.
11) Octet eliminates three variables.

Minimization of SOP expression:-

Minimization procedure:-

1) prepare the k-map & place is according to the given truth table or logical expression. Fill remaining cell 0's.
2) Locate the isolate 1's i.e. the 1's which cannot be combined with any other 1. Encircle such 1's.
3) Identify the 1's which can be combined to form a pair in only one way & encircle them
4) Identify the 1's which can form a quad in only one way & encircle them.

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5) Identify the 1's which can form an octet in only one way & encircle them.

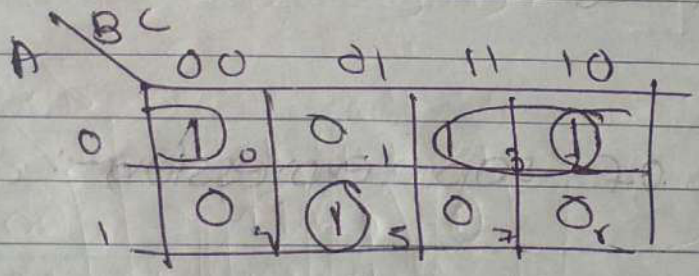
6) After identifying the pairs, quad's & octets check if any 1 is yet to be encircled. If yes encircle them with each other or with the already encircled 1's. (by means of overlapping).

4.8.) Logical expression in the standard SOP form is as follow.

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C$$

minimize it using K-map technique

⇒ prepare K map place 1's & 0's.



$$\boxed{Y = A\bar{B}C + \bar{A}\cdot B + \bar{A}\cdot\bar{C}}$$

F

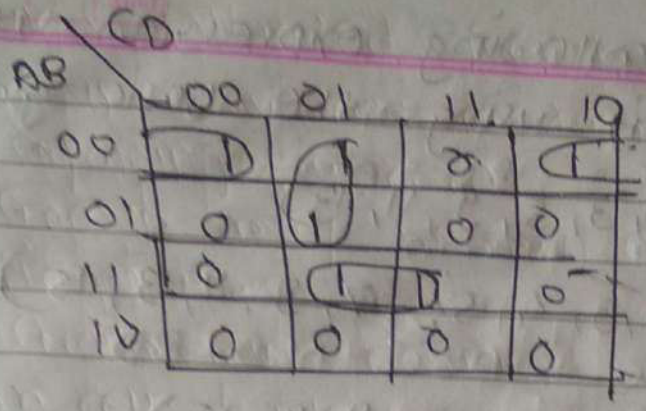
② logical expression represented by logic ckt. is

$Y = \Sigma m(0, 1, 2, 5, 13, 15)$ Draw the K-map & find the minimized logical expression

⇒ $Y = m_0 + m_1 + m_2 + m_5 + m_{13} + m_{15}$

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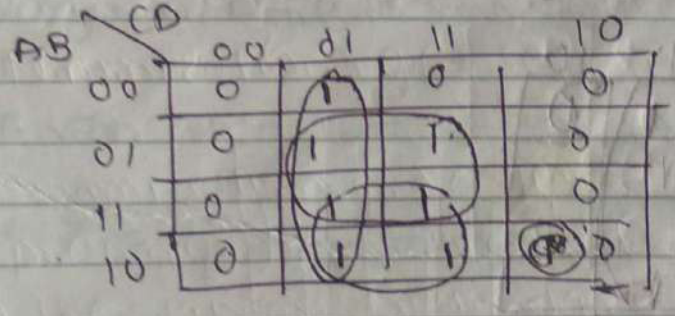
$$Y = \bar{A} \cdot C \cdot D + A\bar{B} \cdot D + \bar{D} \cdot \bar{A}\bar{B}$$

4.8.3 For logical expression given below draw the K-map & obtain the simplified logical expression.

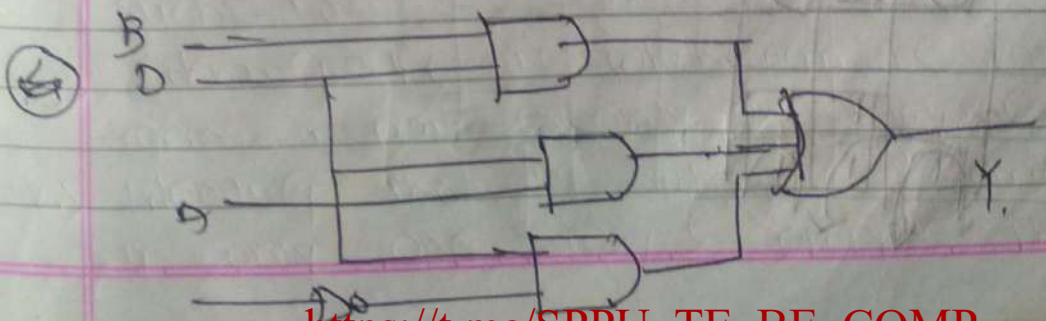
$$Y = \Sigma m (1, 5, 7, 9, 11, 13, 15)$$

Realize the minimized expression using basic gate

→ $Y = \Sigma m$



$$Y = B \cdot D + A \cdot D$$



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④ Minimize the following expression using K-map & realize it using the basic gates.

i) $Y = \Sigma m(1, 3, 5, 9, 11, 13)$

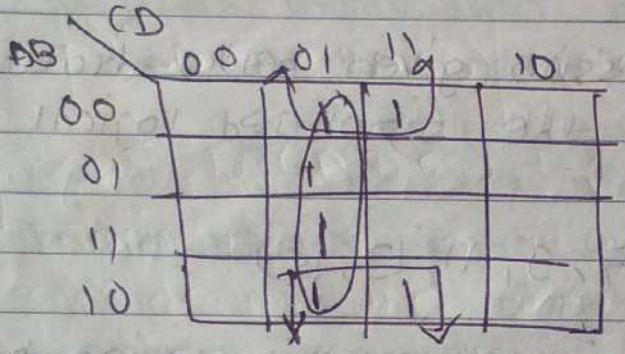
ii) $Y = \Sigma m(1, 3, 4, 5, 7, 9, 11, 13, 15)$

iii) $Y = \Sigma m(1, 2, 9, 10, 11, 14, 15)$

iv) $Y = \Sigma m(0, 2, 5, 6, 7, 8, 10, 13, 15)$

⇒

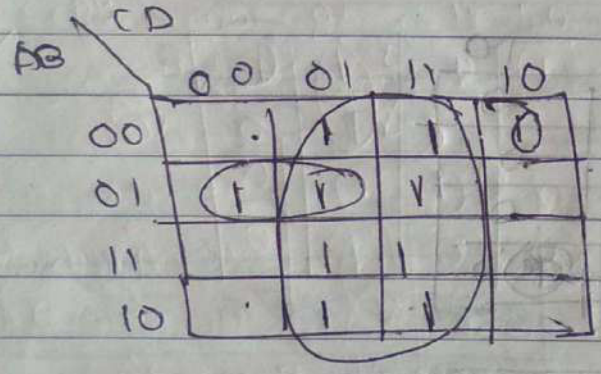
i)



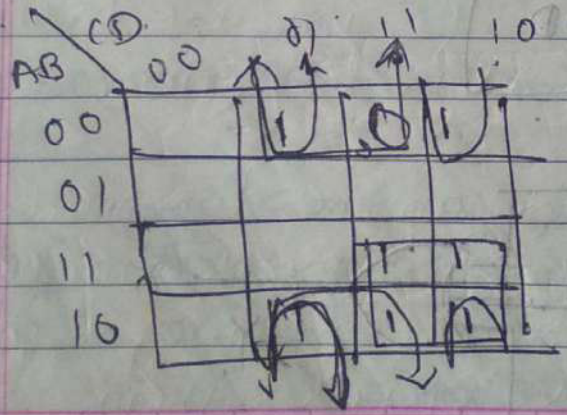
$N = 7 / (2^n - 1)$

$Y = \bar{C}D + \bar{B} \cdot D$

②



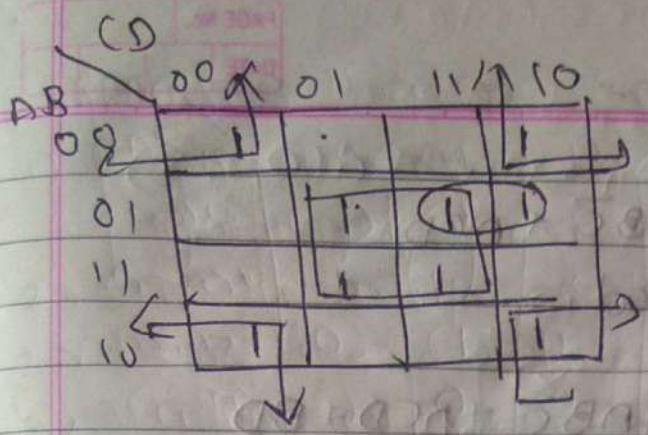
③



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$N \leq 2^n - 1$



$$Y = B \cdot D + \overline{A}B \cdot C + \overline{B}\,\overline{D}$$

(Imp)

Elimination of redundant group:-

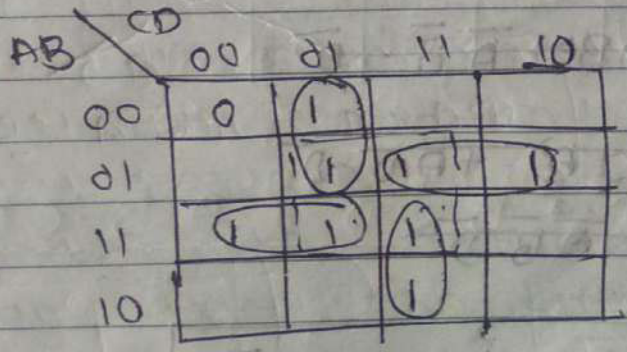
1) If all the 1's in a group are already involved in some other groups, then that group is called redundant group.
2) The redundant group has to be eliminated because it increases the no of gates required

4.8.8 Minimize the following expression using the k-map.

$Y = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$

→ Given exp can be in std form

$$Y = m_1 + m_5 + m_6 + m_7 + m_{11} + m_{12} + m_{13} + m_{15}$$



Quad by dotted line

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Expression without Quad :-

$Y = \bar{A}\bar{C}D + \bar{A}BC + AB\bar{C} + ACD$

Expression with Quad

$Y = \bar{A}\bar{C}D + \bar{A}BC + AB\bar{C} + ACD + BD$

quad should not encircle it is called as redundant group.

$Y = D(AC + \bar{A}\bar{C}) + (\bar{A}C + A\bar{C})B$

$$\boxed{Y = D(\overline{A \oplus C}) + (A \oplus C) \cdot B}$$

Y

② minimize the following expression using K-map

$Y = \Sigma m(0, 1, 2, 5, 13, 15)$

⇒

AB \ CD	00	01	11	10
00	1	1	0	1
01	0	1	0	0
11	0	1	1	0
10	0	0	0	0

Redundant group.

$Y = \bar{A}B + ABD + \bar{A}\bar{B} \cdot \bar{D}$

$= B(A + \bar{A}) + \bar{A}\bar{B} \cdot \bar{D}$

$Y = B + \bar{A}\bar{B} \cdot \bar{D}$

$$\boxed{Y = \bar{A}\bar{C}D + ABD + \bar{A}\bar{B}\bar{D}}$$

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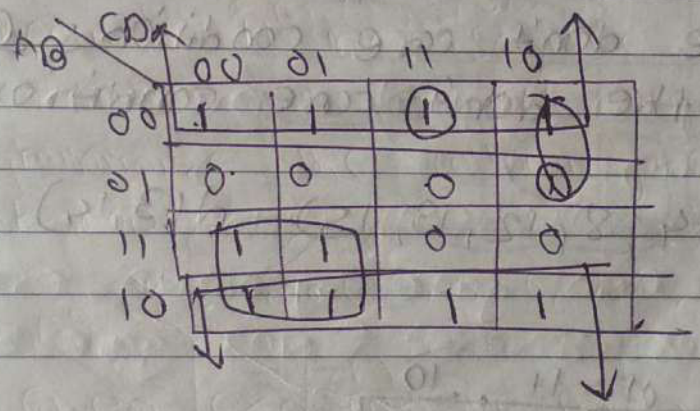
minimization of logic functions not specifie in sop form:-

1) Enter 1's for minterm
2) Enter a pair of 1's for each term with one less variable than total ($\bar{A}B\bar{C}$, $\bar{A}\bar{B}\bar{D}$)
3) Enter a four adjesant one for each term with two less variable than total. (i.e. $A\bar{C}$)
4) Repeat process. Once k-map obtained the procedure is same as earlier.

4.8.10 minimize the logic equation given below.

$Y = \bar{A}\bar{B}CD + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{D} + B\bar{C} + \bar{B}$

⇒



$Y = A\bar{C} + \bar{B} + \bar{A}\bar{C}\bar{D}$

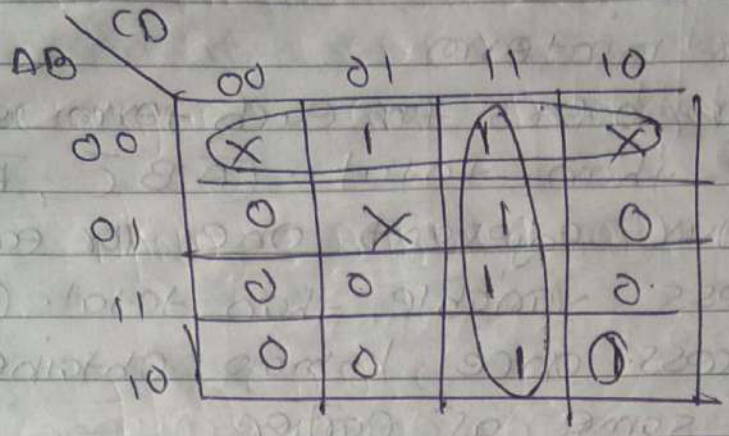
Don't care conditions:- x (0 or 1)
Don't care condition x (0 or 1)
d ( )

4.8.11 simplify the expression given below using k-map. The don't care condition are indicated by d ( ).

⇒ $Y = \Sigma m(1, 3, 7, 11, 15) + d(0, 2, 5)$

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$Y = m_1 + m_3 + m_7 + m_{11} + m_{15} + d(0,2,5)$

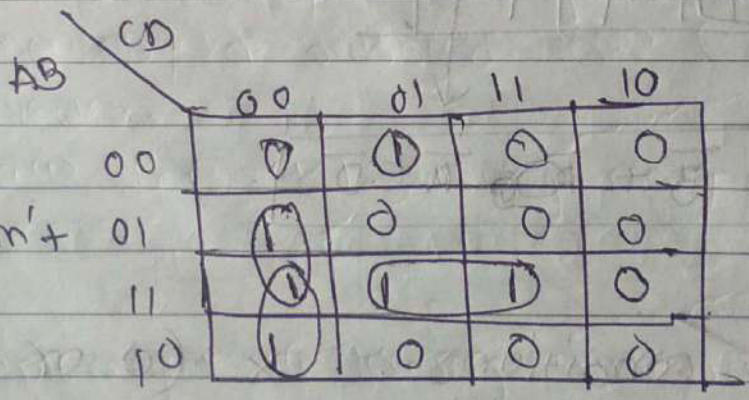


$Y = \bar{A}\,\bar{B} + CD$ (Even don't care need not to consider while grouping)

(12) Simplify the boolean expression once by considering the don't care conditions & once by ignoring the don't care condition.

$Y = \Sigma m(1,4,8,12,13,15) + d(3,14)$

⇒



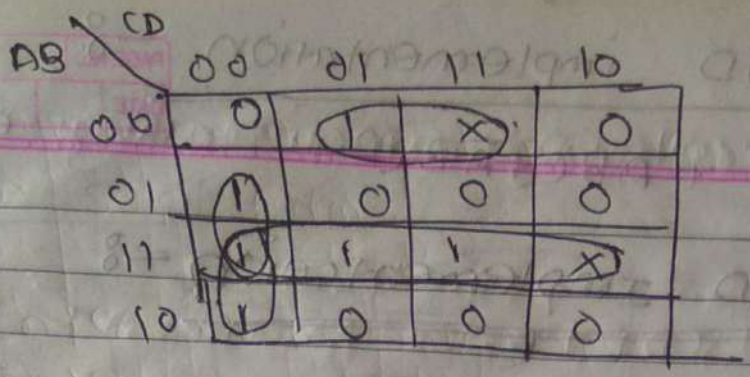
without don't care

$Y = A\cdot\bar{C}\bar{D} + B\cdot\bar{C}\cdot\bar{D} + \bar{A}\cdot\bar{B}\cdot\bar{C}D + ABD$

with don't care

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$Y = \bar{A}\bar{B}\cdot D + AB + \bar{C}\cdot\bar{D}\cdot B + \bar{C}\cdot\bar{D}A$

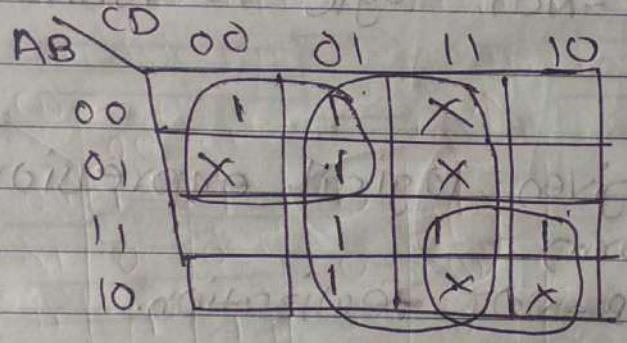
$Y = AB + \bar{C}\cdot\bar{D}(A+B) + \bar{A}\cdot\bar{B}D$

It reduces the no. of gate

(13) Minimize the following expression using k-map.

$Y = \Sigma m(0,1,5,9,13,14,15) + d(3,4,7,10,11)$

⇒



$Y = \bar{A}\cdot\bar{C} + D + A\cdot C$

$= \overline{(A \oplus C)} + D$

Disadvantage of K-map:-

1) We can't use computer for k-map solution
2) minimization is complicated no. of variable exceed 6.
3) Simplification process depend on user.

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Product of Sum (POS) Method

1) POS term → Maxterm.
2) Represent 0 for each maxterm
3) K-map corresponding to POS form

A \ B	0	1
0	$A+B$	$A+\overline{B}$
1	$\overline{A}+B$	$\overline{A}+\overline{B}$

(a) 2-variable K-map.

A \ BC	00	01	11	10
0	$A+B+C$	$A+B+\overline{C}$	$A+\overline{B}+\overline{C}$	$A+\overline{B}+C$
1	$\overline{A}+B+C$	$\overline{A}+B+\overline{C}$	$\overline{A}+\overline{B}+\overline{C}$	$\overline{A}+\overline{B}+C$

(b) 3-Variable K-map. M_0, M_1

* Representation of Standard POS form on K-map :-
1) Logical expression in the Std. POS form can be represent on K-map by entering 0's in the cell's of K-map corresponding to each term present in the given eqn.
2) The remaining cell's are filled with 1's.

Eg

1) Represent the following standard POS expression on Karnaugh-map.

⇒ i) $Y = (A+B+C)(A+\overline{B}+C)(\overline{A}+\overline{B}+C)$

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⇒

A \ B	00	01	11	10
0	0	1	1	0
1	1	1	1	0

enter 0's

ii) $Y = (A+B+C+\overline{D})\cdot(A+B+\overline{C}+\overline{D})\,(A+\overline{B}+\overline{C}+\overline{D})\,(\overline{A}+\overline{B}+C+D)$

↳

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	0	1
11	0	1	1	1
10	1	1	1	1

* simplification of standard pos form using k-map :-

Minimization procedure :-

1) The given pos expression consists of maxterms
2) corresponding to every maxterm enter a 0 in the k-map.
3) Enter 1's in the remaining cells of k-map.
4) Encircle group of 0's instead of 1's for carrying out the simplification.
5) Rules of simplification are exactly same those for the sop form.

4.10.3 Find the expression in the sop form for the k-map.

A \ BC	00	01	11	10
0	1	1	0	0
1	1	1	0	1

⇒ $Y = (A+\overline{B})\,(\overline{B}+\overline{C})$

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4.10.4 minimize the following standard POS expression using K-map.

$$Y = \pi M(0, 2, 3, 5, 7)$$

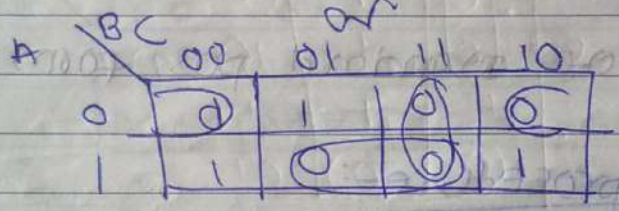
⇒



~~$Y = (\bar{A} + \bar{B})(\bar{A} + \bar{B})(A +$~~

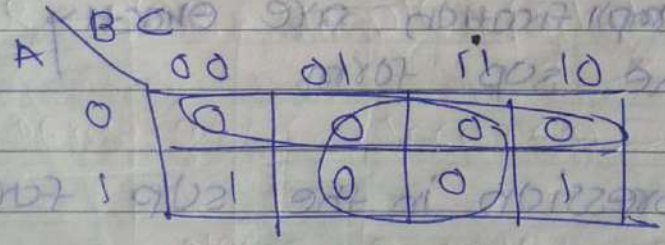
$$Y = (\bar{B} + \bar{C})(A + \bar{B})(\bar{A} + C)$$

or



$$\boxed{Y = (B + C)(\bar{A} + \bar{C})(\bar{A} + C)}$$

(5) Find the expression for the O/p in the POS form for the K-map.



⇒

$$Y = A \cdot (\bar{C})$$

$$\boxed{Y = A \cdot \bar{C}}$$

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(6) write the simplified expression for o/p in the pos form for the following expression

$Y = \pi M (0, 2, 3, 7)$

$\Rightarrow$

$Y = M_0 \cdot M_2 \cdot M_3 \cdot M_7$

A \ BC	00	01	11	10
0	0	1	0	0
1	1	1	0	1

$Y = (A + C)(\overline{B} + \overline{C})$

(7) write the expression for o/p in the pos form for the k-map

$\Rightarrow$

AB \ CD	00	01	11	10
00	0	1	1	0
01	1	1	1	1
11	1	1	1	1
10	0	1	1	0

$Y = B + D$

(8) For the k-map write the expression for o/p in the pos form

AB \ CD	00	01	11	10
00	0	1	1	1
01	0	0	0	1
11	1	1	0	0
10	1	1	0	0

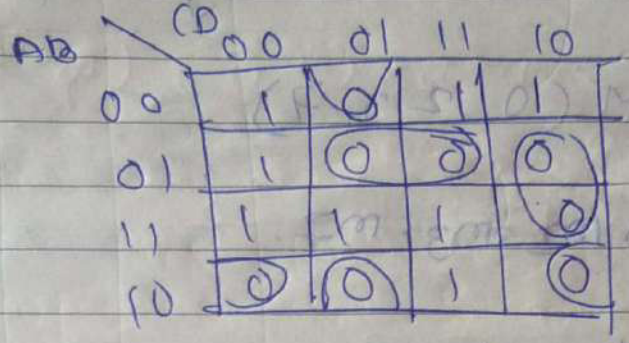
$Y = (\overline{C} + D + A)(\overline{B} + A + \overline{B} + \overline{D})(\overline{A} + \overline{C})$

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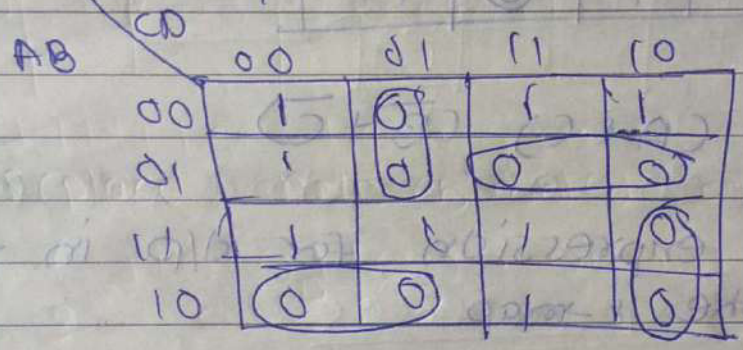
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(9) Obtain the expression for o/p in the POS form.

⇒



Another way :-



$Y = (A + C + \bar{D}) \cdot (A + \bar{B} + \bar{C})$

$(\bar{A} + B + C)\ (\bar{A} + \bar{B} + C)$

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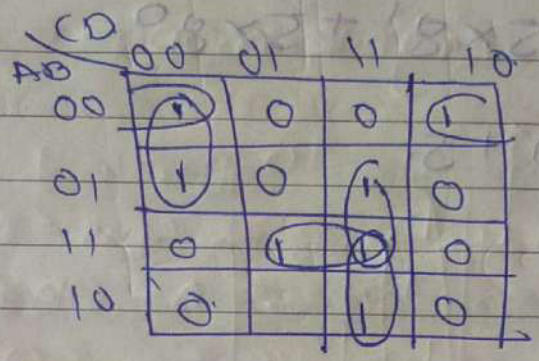
(3) solve the following using minimization technique.

i) $Z = f(A, B, C, D) = \Sigma(0, 2, 4, 7, 11, 13, 15)$

ii) $Z = f(A, B, C, D) = \Pi(1, 2, 3, 6, 8, 11, 14, 15)$

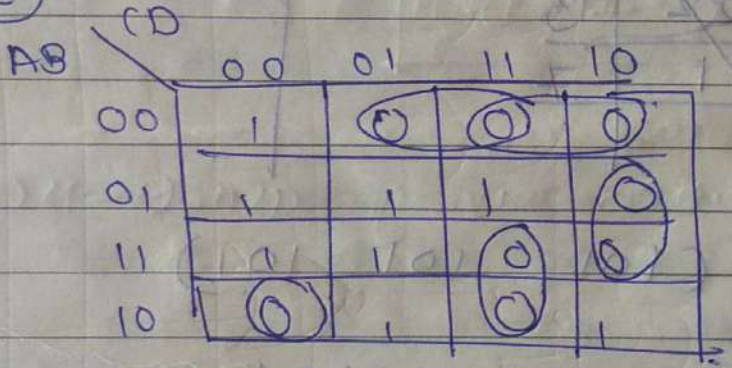
$\Rightarrow$

(1)



$Y = \bar{A}\bar{B}\bar{D} + \bar{A}C\bar{D} + BCD + ABD + ACD$

(2)



$Y = (A + B + \bar{D})(A + B + \bar{C})(\bar{B} + \bar{C} + D)$

$(\bar{A} + \bar{C} + \bar{D})(\bar{A} + B + C + D)$

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* Signed Binary Numbers :-

Sign Magnitude Representation :-

i) In the decimal number system a (+) plus sign is used to denote a positive number & a minus (−) for denoting a negative number.

ii) plus sign usually dropped.

iii) Digital circuits can understand only two symbols 0 & 1. we must use some symbol to indicate sign of the number also.

iv) An additional bit is used as the sign bit & it is placed at Most Significant bit (MSB).

v) A 0 is used to represent a positive number & a 1 to represent a negative number.

e.g :- 8-bit signed number

i) $(01000100)_2$ Represent positive no.

MSB indicates no. is +ve

$(01000100)_2 = (68)_{10}$

ii) $(11000100)_2$ represent negative no.

MSB indicates no is −ve

$(11000100)_2 = -(68)_{10}$

the 1 in the MSB indicates that the number is −ve & other seven bits gives magnitude. This kind of representation known as "Signed-Magnitude Representation".

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Example-:

Find the decimal equivalent of the following binary numbers assuming sign magnitude representation of the binary numbers.

a) 101100
b) 001000
c) 0111
d) 1111

⇒ a) sign bit is 1 means no is -ve

$(101100)_2 = (-12)_{10}$

b) Sign bit is 0 means no. is +ve

$(001000)_2 = (+8)_{10}$

c) $(0111)_2 = (+7)_{10}$

d) $(1111)_2 = (-7)_{10}$

One's complement Representation:

i) In a binary number, if each 1 is replaced by 0 & each 0 by 1. The resulting number is known as one's complement of first number.

ii) Both the numbers are complement of each other.

iii) If one of these number is positive then the other number will be negative with same magnitude & vice versa.

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iv) this method is widely used for representing signed numbers.

v) In this representation MSB is 0 for positive numbers & 1 for negative numbers.

Example -:

* Find the one's complement of the following binary numbers.

a) 0100 1110 01
b) 1101 1010

⇒

a) 1011000110

b) 00100101

Example -:

* Represent the following numbers in one's complement form

a) +7 and -7
b) +8 and -8
c) +15 and -15

⇒ In one's complement representation

a) $(+7)_{10} = (0111)_2$ & $(-7)_{10} = (1000)$

b) $(+8)_{10} = (01000)_2$ & $(-8)_{10} = (10111)_2$

c) $(+15)_{10} = (01111)_2$ & $(-15)_{10} = (10000)_2$

It is observed that for n bit numbe

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The maximum positive number which can be represented in 1's complement representation $(2^{n-1}-1)$ & maximum negative number $-(2^{n-1}-1)$.

Two's complement Representation:-

i) If 1 is added to 1's complement of a binary number is known as the two's complement of the binary numbe
e.g:- 2's complement of 0101 is 1011. since 0101 represents $(+5)_{10}$ therefore $(-5)_{10}$ in 2's complement representation.

ii) If the MSB is 0 then number is positive. MSB is 1 the number is negative.

iii) For an n-bit number, the maximum positive number which can be represe-nted in 2's complement form is $(2^{n-1}-1)$ & the maximum negative no. is -2^{n-1}

* Example:-
Find the 2's complement of the numbers.
i) 01001110 ii) 00110101

⇒

i) Number 01001110
is comp 10110001
Add 1 +1
10110010

ii) 00110101
is comp: 11001010
Add 1 +1
11001011

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Sign Magnitude is & 2's complement Representation using four bits.

Decimal no.	Binary number Sign Magnitude	1's comp of signed no.	2's comp of signed no.
0	0000	0000	0000
1	0001	0001	0001
2	0010	0010	0010
3	0011	0011	0011
4	0100	0100	0100
5	0101	0101	0101
6	0110	0110	0110
7	0111	0111	0111
-8	-	-	1000
-7	1111	1000	1001
-6	1110	1001	1010
-5	1101	1010	1011
-4	1100	1011	1100
-3	1011	1100	1101
-2	1010	1101	1110
-1	1001	1110	1111
-0	1000	1111	-

From the table, it is observed that the maximum positive number is 0111 = +7 whereas the maximum negative number is 1000 = -8 using four bit in 2's complement format.

Example:-
Represent $(-17)_{10}$ in
i) Sign Magnitude
ii) one's complement
iii) Two's complement.

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⇒ The minimum no. of bits required to represent $(+17)_{10}$ in signed no. format is six

$$(+17)_{10} = (010001)_2$$

Therefore $(-17)_{10}$ is represented by

i) 110001 in sign Magnitude representation.
ii) 101110 in 1's complement form.
iii) 101111 in 2's complement form.

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